CAMPUS CONNECT : A UNIVERSITY-VERIFIED MARKETPLACE FOR BUYING, SELLING, RENTING AND EXCHANGING STUDENT RESOURCES



CENTRAL UNIVERSITY OF KARNATAKA

A Project Submitted in Partial Fulfillment of the Requirements in the Subject of Advance Software Engineering in the Degree of

MASTER OF COMPUTER APPLICATIONS

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November, 2026

Campus Connect: A University-Verified Marketplace for Buying, Selling, Renting and Exchanging Student Resources

***Abstract*—**Students routinely buy, sell, rent, and discard textbooks, furniture, electronics, and other study-related items, yet general-purpose marketplaces suggestion neither verification of campus affiliation nor features adapted to campus life. This paper presents Campus Connect, a university-verified marketplace in which students, faculty, and staff can buy, sell, rent, and exchange items within their own institution. The motivation is to combine the convenience of online listing with the trust associated with a bounded, identifiable community, while encouraging the reuse of student resources. The proposed system comprises university e-mail authentication, snap listings, group browsing and filtering, four transaction modes, in-app chat, payment selections, ratings and reviews, a Wishlist, and notifications, together with campus event notifications and off-campus errand requests. The paper specifies the system using the organization of IEEE Std 830-1998 and derives a layered architecture, use cases, a logical data model, a security and privacy analysis, a development methodology, and a proposed test framework. A web minimum viable product (HTML, CSS, JavaScript, React, MongoDB) is described in the project material; however, no practical results are available, so all benefits are presented as expected outcomes to be validated. The principal contribution is a requirements-driven design of a campus-restricted marketplace together with an assessment plan. Institutional verification is identified as the central design element, and directions for future research are outlined.

***Keywords*—**campus marketplace, second-hand trading, user verification, trust and reputation, software necessities requirement, IEEE Std 830-1998, peer-to-peer commerce, circular economy.

I. INTRODUCTION

Second-hand trading among university students usually takes place through social media groups, notice boards, or general classified services. These channels are convenient but expose participants to strangers, provide no assurance of campus affiliation, and offer little support for organizing items by student needs. Prior work on local second-hand trading platforms has examined how negotiation costs shape transaction satisfaction [1], which indicates that the context in which peers meet and negotiate matters for the quality of the transaction.

General marketplaces are open to any registrant. Consequently, a student who wishes to sell a textbook or rent a bike to another student must filter listings from a very large and anonymous population and must often arrange a meeting with someone whose identity cannot be check. Safety, trust, and wasted search effort are therefore recurring concerns. At the same time, university campuses generate predictable cycles of reusable goods: textbooks at the end of a semester, furniture and appliances at the end of an academic year, and laboratory equipment or notes that remain useful to junior cohorts.

Empirical studies of collaborative consumption report motivations such as sustainability, enjoyment, and economic benefit [2], and conceptual work on the circular economy commonly characterizes it through reduce, reuse, and recycle activities [3]. A campus-specific marketplace can support such reuse at the scale of one institution, where the pool of participants is bounded and their affiliation can be checked.

This paper presents the design of **Campus Connect**, a university-verified marketplace for buying, selling, renting, and exchanging items among students, faculty, and staff. Throughout the paper each capability is labelled as **described** (stated in the project presentation), **proposed** (a design choice introduced in this paper to make the described capability specifiable), or **conceptual/future** (stated as a possible extension but not specified). The paper makes the following contributions:

1) A formal statement of the research problem and research questions for campus-restricted marketplaces (Sections II–III).
2) A system concept that distinguishes described, proposed, and future capabilities (Section IV).
3) A requirements specification organized according to IEEE Std 830-1998, with functional and non-functional requirements (Section V).
4) An architecture, use-case model, logical data model, and security analysis (Sections VI–IX).
5) A development methodology and a proposed test framework (Sections XII–XIII). No empirical evaluation is reported.

II. PROBLEM STATEMENT

The research problem is to design an online marketplace that is restricted to the members of a single university and that supports multiple transaction modes while providing mechanisms for trust and communication, in a form that can be specified, implemented, and verified. The following sub-problems are identified in the project material and in the literature reviewed in Section III.

P1. Lack of a trusted student-only marketplace. Existing channels mix campus members with outsiders. **P2. Unverified users.** Anyone may register on a general platform, so a counterpart's affiliation cannot be checked. **P3. Discovery within one campus.** It is difficult for a student to find buyers or sellers on the same campus without filtering a large, unrelated listing population. **P4. Safety.** Meeting unknown individuals to complete a trade creates personal-safety concerns. **P5. Underutilization of resources.** Reusable textbooks, furniture, and equipment are discarded because no convenient route exists to pass them on. **P6. Fragmented communication.** Negotiation takes place on separate messaging applications, which are disconnected from the listing and from any record of the transaction.

Accordingly, three research questions are posed. **RQ1:** What functional and non-functional requirements should a campus-restricted marketplace satisfy? **RQ2:** How can institutional verification be combined with ratings and reviews to establish trust? **RQ3:** What architecture and verification approach are appropriate for such a system?

III. RELATED WORK

A. Trust and Reputation in Online Marketplaces

Resnick and Zeckhauser looked at the eBay feedback system and found that feedback was given very frequently, was almost invariably positive and did have predictive power (though the net score displayed was not the best predictor [4]). Jøsang et al. looked at trust and reputation systems in the context of online service provision [5]. Pavlou and Gefen looked at the role of institutionally based mechanisms (i.e. feedback and escrow) in the building of trust in online marketplaces [6]. Taken together, these works motivate reputation mechanisms but also indicate that ratings alone are an imperfect trust signal.

B. Local and Campus Marketplaces

Park et al. studied local second-hand trading platforms from the perspective of negotiation costs [1]. Wu and Nie argue that the general information-issuing platforms are poorly directed and disorganised in terms of meeting student needs, and propose a campus second-hand trading website [7]. Zeng et al. collected function points through user research among college students and designed an application for second-hand and virtual resource trading [8]. Two student project reports are also relevant. UKMThrift [9] is a PHP/MySQL web application. a Malaysian university platform that validates matriculation numbers, has administrator verification of users and products, and provides chat, wishlist and status management, with payment integration and mobile applications out of scope. ReTrade [10] is an application for an Indian university, built with React Native and Firebase. The application relies on the closed-community premise for trust. Buyers can interact with sellers via chat or telephone. Online payment is listed as future scope. Both reports are unreviewed and are provided for design reference only, not as validated evidence.*Security, Pricing, and Recommendation*

C. OWASP Application Security Verification Standard presents security test criteria for web applications [11] and is used as a source of information for non-functional and security testing. In order to help with the problem of pricing, Pudaruth used machine learning methods for predicting prices for used cars [12], and Han et al. suggested a multimodal price suggestion system for used car listings based on visual and textual information [13]. General methods of recommendation are described in [14]. These references will be useful only for future work (Section XVI). *Comparison and Research Gap*

TABLE I
COMPARISON OF RELATED RESEARCH AND PLATFORMS

Research / platform	Main approach	Verification	Marketplace	Trust mechanism	Limitation
Resnick & Zeckhauser [4]	Empirical analysis of eBay feedback	Not addressed	General online auction	Bilateral feedback	Feedback skew; not campus-specific
Pavlou & Gefen [6]	Study of institution-based trust	Not addressed	General online marketplace	Feedback, escrow-type mechanisms	No campus context; no system specification
Park et al. [1]	Study of local second-hand platforms	Not the focus	Local second-hand	Neighborhood context	Behavioral study; no system design
Wu & Nie [7]	Design of campus trading website	Not stated in abstract	Campus second-hand	Not stated in abstract	Abstract-level information only
Zeng et al. [8]	User-research-driven app design	Not stated in abstract	Second-hand and virtual resources	Not stated in abstract	Abstract-level information only
UKMThrift [9]	PHP/MySQL web platform, MVC	Matric-number validation, admin approval	Campus second-hand (one university)	Feedback, admin moderation	Payment and mobile out of scope; self-reported tests
ReTrade [10]	React Native + Firebase app	Closed-community premise; no institutional check described	Campus second-hand (one university)	Chat; offline contact	No payment; no verification described
Campus Connect (this paper)	Requirements-driven design; web MVP and mobile app	University e-mail verification	Buy, sell, rent, exchange; events; errands	Verified community, ratings, in-app chat	Proposed; not yet evaluated

The surveyed sources treat trust in general marketplaces [4]–[6], campus trading functionality [1], [7]–[10], and price prediction [12], [13] largely in isolation. Within the sources reviewed, none was found that combines (i) institution-verified membership, (ii) four transaction modes including renting and exchange, (iii) campus-life services such as event alerts and errand requests, and (iv) a requirements specification organized according to a recognized standard. This is the gap addressed by the present paper. The review is not exhaustive and the gap statement is limited to the sources examined.

IV. PROPOSED SYSTEM

A. System Overview

Campus Connect is described in the project presentation as a university-verified platform where students, faculty, and staff buy, sell, rent, and exchange items within their own campus community. Two delivery channels are described: a mobile application (Android and iOS) as the primary experience and a web minimum viable product (MVP) built with HTML, CSS, JavaScript, React, and MongoDB. The described usage flow has six steps: sign up with a university email address, get verified, browse or post a listing, chat with the buyer or seller, meet on campus or pay in-app, and rate the exchange (Fig. 1 and Fig. 2).

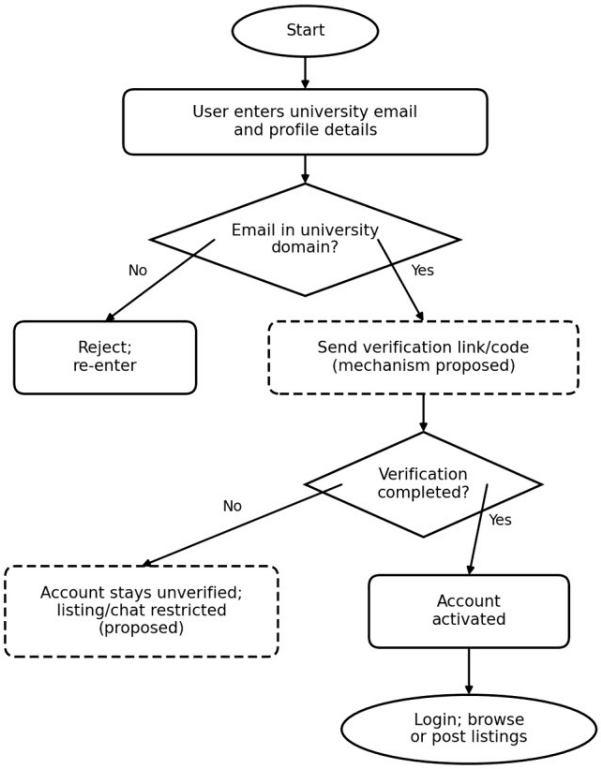


Fig. 1. User registration and verification workflow.

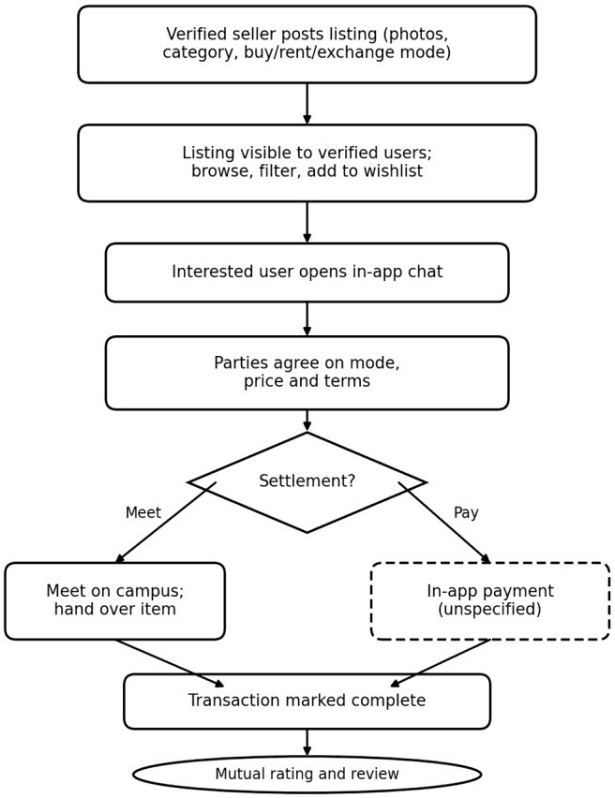


Fig. 2. Product listing and transaction workflow.

B. Target Users

The described users are students, faculty, and staff of the university. Alumni appear in the described alumni and placement assistance feature, where verified alumni post job and internship referrals and chat with students; alumni are therefore treated as a user class where applicable. An administrator or moderator role is **proposed** to support verification and moderation.

C. Core Functionalities

University verification and access (described; mechanism proposed). Users sign up with a university email address and are verified. The verification mechanism is not specified in the source; a domain check followed by a confirmation link or one-time code is proposed (Fig. 1). **Listings, images, browsing, and filtering (described).** Users post listings with photographs and browse and filter them by category; keyword search is a proposed refinement. **Buy, sell, rent, exchange (described).** A listing carries one transaction mode. **In-app chat (described)** links a conversation to a listing. **Payment options (described).** The presentation lists flexible payment options and describes settlement either by meeting on campus or by paying in-app; the payment mechanism is not specified. **Ratings and reviews, wishlist, and notifications (described).**

D. Additional Campus Features

Campus event notifications (described): alerts for fests, flea markets, swap meets, and sales on campus. **Off-campus errand requests (described):** a student asks a fellow student who is already travelling to town to collect items and bring them back. **Alumni and placement assistance (described):** referrals and direct alumni chat. Skill mapping, internship and placement tracking, and an academia–industry collaboration portal are mentioned in the project scope but are not elaborated in the presentation; they are treated as **conceptual** and are excluded from the requirements in Section V.

E. Future Extensions

The presentation lists the following as future directions, none of which is implemented in the described system: expansion to more campuses, AI-assisted price suggestions, a scheduled pickup and delivery network, a sustainability impact tracker, loyalty rewards and university ID integration, and secure in-app escrow payments.

V. REQUIREMENTS SPECIFICATION (IEEE STD 830-1998)

IEEE Std 830-1998 is a recommended practice that describes the content and qualities of a good software requirements specification (SRS) [15]. It is not a formatting standard for research papers; it is used here only to organize the requirements of the proposed system. The standard has since been superseded by ISO/IEC/IEEE 29148 [16], which retains the same basic concerns. The section follows the three top-level parts of the standard: Introduction, Overall Description, and Specific Requirements.

A. Introduction

1) Purpose. This specification defines the requirements of Campus Connect for developers, testers, university stakeholders, and prospective users. **2) Scope.** The product enables verified university members to create listings and to discover, discuss, and complete buy, sell, rent, and exchange transactions, and provides event notifications and errand requests. Delivery logistics, escrow payments, and AI-assisted pricing are outside the current scope. **3) Definitions.** *Verified user*: a user whose university e-mail address has been confirmed. *Listing*: an item offered under one transaction mode. *Transaction mode*: buy/sell, rent, or exchange. *MVP*: minimum viable product.

FR/NFR: functional/non-functional requirement. *UC*: use case. **4) References.** See [15], [16] and the reference list. **5) Overview.** Subsection B gives the overall description and Subsection C the specific requirements. Requirement status is marked D (described in the project presentation), P (proposed here), or C (conceptual).

B. Overall Description

1) Product perspective. Campus Connect is a new, self-contained product that interacts with a university e-mail service for verification (proposed) and, optionally, with a payment provider (conceptual). **2) Product functions.** The functions are those of FR-01 to FR-15 in Table II. **3) User classes.** Students (primary users), faculty and staff, alumni (referral features), and administrators/moderators (proposed). **4) Operating environment.** Current web browsers, Android and iOS devices, and a MongoDB database (described); hosting is unspecified. **5) Design and implementation constraints.** The MVP uses HTML, CSS, JavaScript, React, and MongoDB (described); access is restricted to university members; applicable data-protection law must be respected. **6) User documentation.** An in-app onboarding guide, terms of use, and safe-meetup guidance (proposed). **7) Assumptions and dependencies.** Users hold active university e-mail accounts; the university permits e-mail-domain verification; network connectivity is available; settlement is either in person or via a payment option whose details remain to be determined (TBD).

C. Specific Requirements

External interfaces comprise the mobile and web user interfaces, the database interface, and an e-mail interface for verification; communication is proposed to use HTTPS. Functional requirements are given in Table II, and non-functional requirements in Table III.

TABLE II
FUNCTIONAL REQUIREMENTS

ID (status)	Requirement name	Description	Input	Processing	Output	Priority
FR-01 (D)	University user registration	Allow a prospective member to create an account with a university e-mail address.	Name, university e-mail, password, role	Validate fields; reject non-university domains; create an unverified account.	Account created; verification prompt	High
FR-02 (D; mechanism P)	University e-mail verification	Confirm control of the university mailbox before transactions are enabled.	Verification link or code	Check token validity and expiry; mark user verified.	Verified account or error message	High
FR-03 (P)	User authentication	Authenticate returning users and manage sessions.	E-mail, password	Check credentials; start session; throttle repeated failures.	Authenticated session or error	High
FR-04 (D)	Product listing	Create, edit, and remove listings with photographs.	Title, description, category, mode, price or terms, photos	Validate input; store listing and images; set status.	Published listing	High
FR-05 (D)	Product browsing	Browse available listings by category.	Category, page	Retrieve listings with status “available”.	List of listing summaries	High
FR-06 (D; keyword P)	Search and filtering	Filter by category; search by keyword and attributes (proposed).	Keywords, category, mode, price range	Query and sort the listing collection.	Filtered results	High
FR-07 (D)	Buy / sell / rent / exchange	Record the chosen mode and outcome; support the described payment options.	Listing, mode, agreed terms, settlement choice	Create transaction record; update listing status; settle in person or via a payment option (TBD).	Transaction record; updated status	High
FR-08 (D)	In-app messaging	Enable chat between interested user and owner about a listing.	Message text, recipient, listing	Store and deliver message; track read state.	Conversation view; unread indicator	High
FR-09 (D)	Ratings and reviews	Allow parties to rate each other after a completed transaction.	Score, comment, transaction	Verify participation; store review; update aggregate rating.	Displayed review and rating	Medium
FR-10 (D)	Wishlist	Allow users to save listings of interest.	Listing identifier	Add or remove entry.	Wishlist view	Medium
FR-11 (D)	Notifications	Inform users of messages, wishlist changes, and transaction updates.	System events	Generate and deliver in-app notification.	Notification list	Medium
FR-12 (D)	Campus event notifications	Publish alerts for campus fests, flea markets, swap meets, and sales.	Event details (authorized poster: P)	Validate and publish; notify users.	Event alerts	Medium
FR-13 (D)	Errand requests	Let a user request that another student fetch items from off campus.	Item description, place, deadline	Publish request; accept; update status.	Accepted or completed errand	Low
FR-14 (D; C)	Alumni referrals	Let verified alumni post job and internship referrals and chat with students.	Referral details	Verify alumni; publish; enable chat.	Referral list	Low
FR-15 (P)	Reporting and moderation	Let users report listings or users; let administrators act.	Report reason, target	Log report; administrator review; hide or suspend.	Resolved report	Medium

TABLE III
NON-FUNCTIONAL REQUIREMENTS (PROPOSED)

ID	Category	Requirement	Verification
NFR-01	Security	Only verified users may post or message; passwords stored as salted hashes; data transmitted over HTTPS.	Security tests (Sec. XIII)
NFR-02	Privacy	Collect minimum personal data; contact details shared only through the application.	Inspection; review
NFR-03	Usability	A new user can register and create a first listing without external help.	Usability test

ID	Category	Requirement	Verification
NFR-04	Performance	Listing and search responses within a threshold TBD at design time.	Load test
NFR-05	Reliability	Regular database backups; graceful handling of failures.	Recovery test
NFR-06	Portability	Consistent functions on web, Android, and iOS.	Cross-platform test
NFR-07	Modifiability	Modular components; requirement IDs traceable to design and tests.	Review
NFR-08	Scalability	Support additional universities by separating institutional data.	Design review

VI. SYSTEM ARCHITECTURE

A layered architecture is proposed (Fig. 3). The **client layer** comprises the web MVP and the mobile application (both described). The **application/backend layer** exposes an API for listings, search, transactions, ratings, and events; its implementation technology is not specified in the source and is left open. The **database layer** uses MongoDB (described), whose document model suits listings with embedded photograph references and variable attributes; image files are proposed to be held in separate object storage. The **authentication and verification layer** validates university e-mail addresses and manages sessions (proposed). The **communication layer** delivers chat messages; the delivery technique (polling or push) is left to design, and the UKMThrift report, for instance, uses periodic polling [9]. The **notification layer** generates in-app alerts for messages, wishlist changes, events, and errands (proposed). A **payment component** is conceptual and is limited to the payment options named in the presentation; escrow is a future extension.

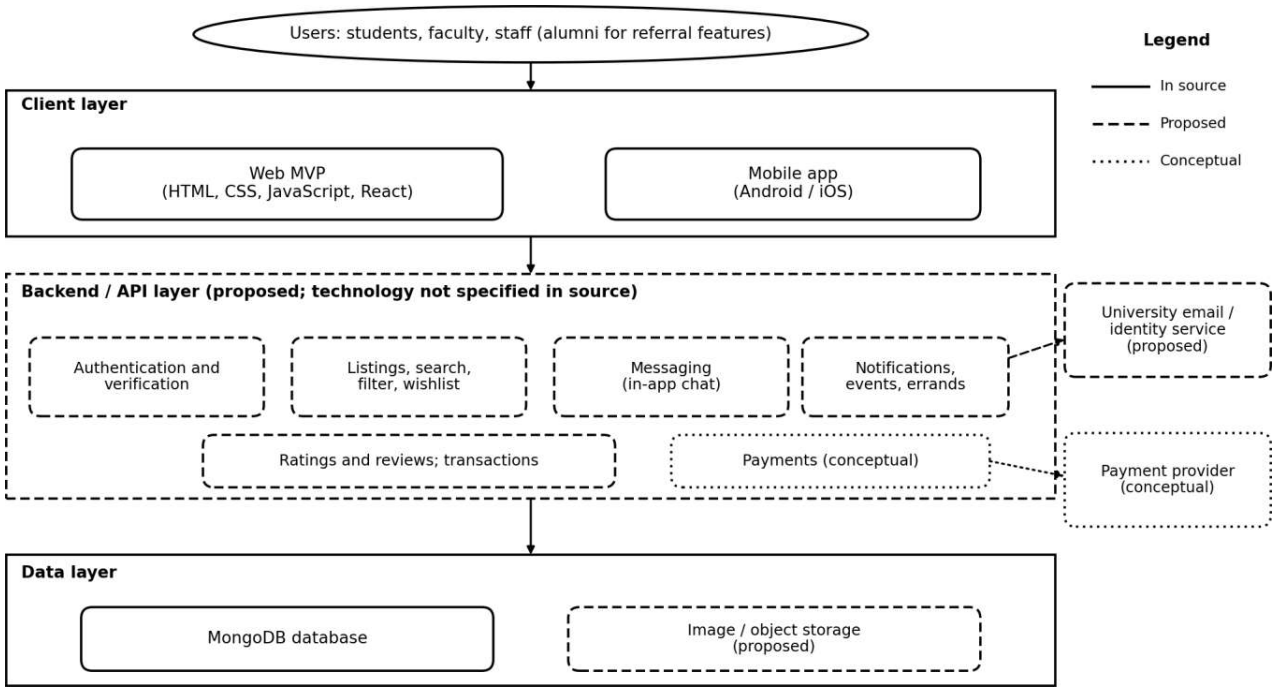


Fig. 3. Proposed layered architecture of Campus Connect. Line style indicates the status of each element.

VII. USE CASE ANALYSIS

Fourteen use cases cover the described functions (Fig. 4 and Table IV). The primary actor is the verified user; the university e-mail service is a secondary actor in verification. Each use case traces to a functional requirement (UC-01→FR-01, UC-02→FR-02, UC-03→FR-03, UC-04→FR-04, UC-05→FR-05/06, UC-06 to UC-08→FR-07, UC-09→FR-08, UC-10→FR-09, UC-11→FR-10, UC-12→FR-11, UC-13→FR-12, UC-14→FR-13).

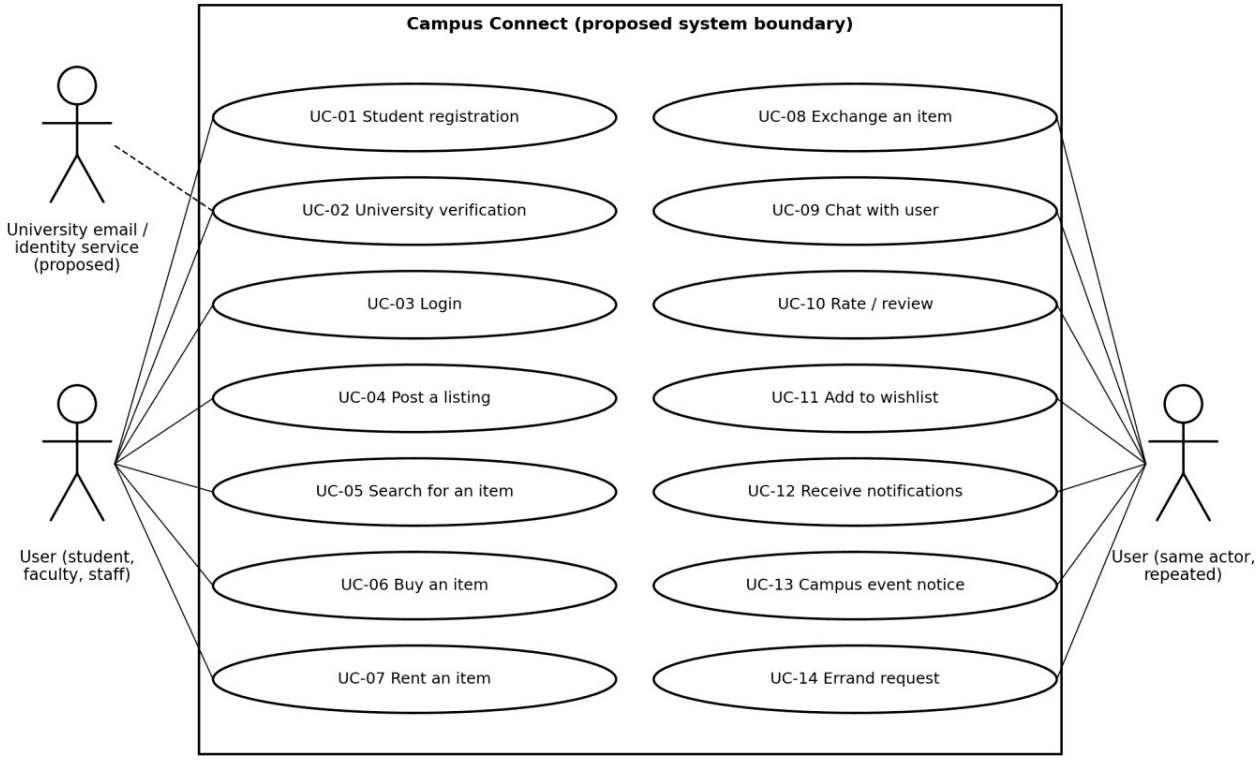


Fig. 4. Use case diagram of Campus Connect (proposed model based on described functions).

TABLE IV
USE CASE SUMMARY

ID	Use case	Actor	Preconditions	Main flow	Alternative flow	Postconditions	Req.
UC-01	Student registration	Prospective user	App or web access	Enter name, university e-mail, password; system validates and creates account.	Non-university e-mail: rejected with message.	Unverified account exists	FR-01
UC-02	University verification	User; e-mail service	Account created	System sends link/code; user submits it; account marked verified.	Expired code: new code issued.	Account verified	FR-02
UC-03	Login	User	Account verified	Enter credentials; system starts session.	Wrong credentials: error; repeated failures throttled.	Authenticated session	FR-03
UC-04	Post a listing	Verified user	Logged in	Enter details, mode, price/terms; attach photos; publish.	Missing fields: validation error.	Listing visible	FR-04
UC-05	Search for an item	Verified user	Logged in	Browse category or enter keywords; apply filters; view results.	No results: suggestion to broaden filters.	Results displayed	FR-05, 06
UC-06	Buy an item	Buyer	Listing available	Open listing; chat; agree price; settle; mark complete.	Item reserved/sold: buyer informed.	Transaction recorded	FR-07
UC-07	Rent an item	Renter	Listing offered for rent	Open listing; agree period and fee; hand over; return; mark complete.	Dates unavailable: choose other dates.	Rental recorded	FR-07
UC-08	Exchange an item	Two users	Listing offered for exchange	Propose swap; owner accepts; exchange on campus.	Owner declines: proposal closed.	Exchange recorded	FR-07
UC-09	Chat with user	Verified user	Listing exists	Open chat from listing; send and read messages.	Recipient offline: message stored.	Conversation saved	FR-08
UC-10	Rate / review	Transaction party	Transaction complete	Enter score and comment; submit.	Not a participant: rejected.	Review stored	FR-09
UC-11	Add to wishlist	Verified user	Logged in	Select listing; add to wishlist.	Already saved: no change.	Wishlist updated	FR-10
UC-12	Receive notifications	Verified user	Events occurred	System generates alert; user opens notification list.	Notifications disabled: no alert.	Alert marked read	FR-11
UC-13	Campus event notification	User; poster	Event published	Authorized poster publishes; users receive alert.	Invalid event data: rejected.	Event visible	FR-12
UC-14	Errand request	Requester; helper	Both verified	Requester posts errand; helper accepts; fetches; hands over.	No helper: request expires.	Errand closed	FR-13

VIII. DATA AND DATABASE DESIGN

The source states that the web MVP uses MongoDB, a document database; the following logical model is therefore **proposed** and has not been reported as implemented. Ten entities are identified (Fig. 5, Table V). A user posts many listings; a category classifies many listings; a listing may be involved in many transactions over time (for example, repeated rentals); each transaction can carry up to two reviews, one per party; users exchange many messages; the wishlist resolves the many-to-many relation between users and listings; notifications belong to a user and may reference an event; and an errand request links a requester and a helper. In a document-oriented realization, photographs would be embedded as an array within the listing and other relations stored as identifier references.

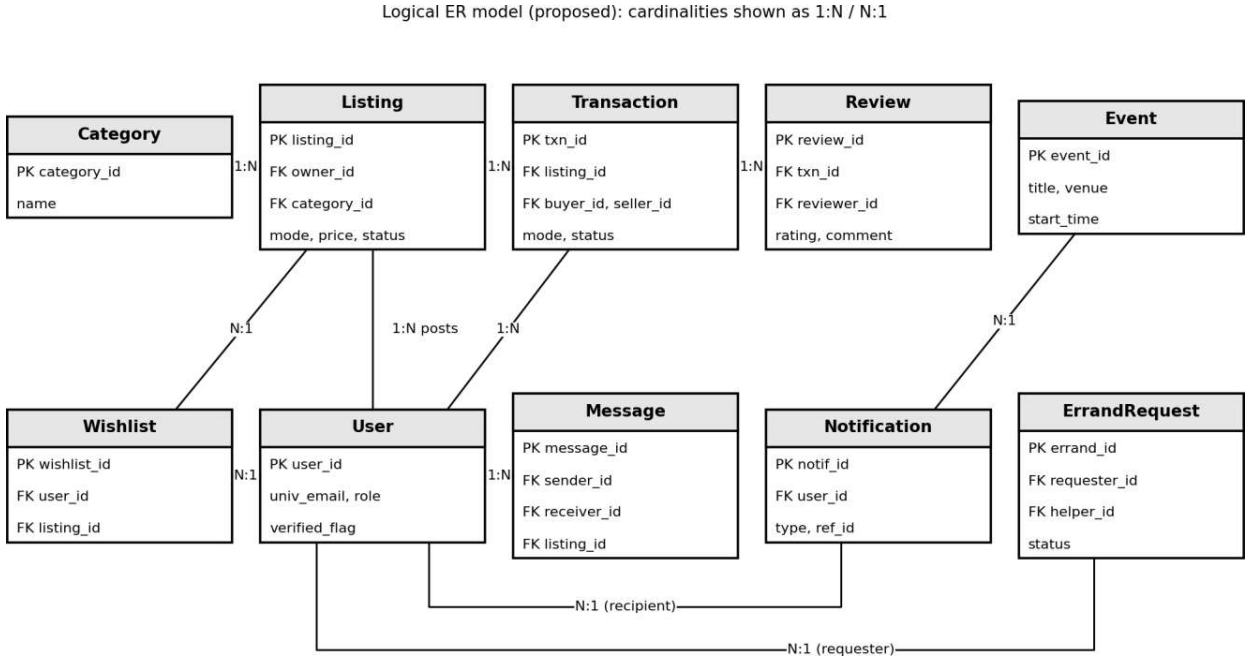


Fig. 5. Proposed logical entity-relationship model (not reported as implemented).

TABLE V
PROPOSED ENTITIES AND ATTRIBUTES

Entity	Important attributes	Purpose
User	user_id, name, university_email, role, verified_flag, rating_summary	Identity, role, and verification state
Listing	listing_id, owner_id, category_id, title, description, mode, price/terms, photos[], status, created_at	Item offered for sale, rent, or exchange
Category	category_id, name	Student-oriented classification and filtering
Transaction	txn_id, listing_id, buyer_id, seller_id, mode, settlement, status, timestamps	Record of an agreed and completed trade
Message	message_id, sender_id, receiver_id, listing_id, text, sent_at, read_flag	In-app chat linked to a listing
Review	review_id, txn_id, reviewer_id, reviewee_id, rating, comment	Reputation signal after a transaction
Wishlist	wishlist_id, user_id, listing_id	Saved listings of interest
Notification	notif_id, user_id, type, reference_id, read_flag	Alerts about messages, wishlist, events, errands
Event	event_id, title, venue, start_time, posted_by	Campus event alerts
ErrandRequest	errand_id, requester_id, helper_id, description, place, deadline, status	Off-campus errand coordination

IX. SECURITY, TRUST AND PRIVACY

University e-mail verification shows that a user controls a mailbox in the university domain; it does not, by itself, prove real-world identity, and it can be circumvented by shared or compromised mailboxes. It is therefore combined with authentication, authorization, reputation, reporting, and moderation. Ratings alone are an imperfect trust signal because feedback tends to be overwhelmingly positive [4]. The OWASP ASVS [11] is proposed as the checklist for the security test cases of Section XIII. Table VI lists principal threats and proposed mitigations; all are design proposals and not reported implementations.

TABLE VI
THREATS AND PROPOSED MITIGATIONS

Threat	Proposed mitigation
Outsiders or fake accounts	Domain check and e-mail confirmation; restrict posting and chat to verified users.
Account takeover	Salted password hashing; session expiry; throttling of failed logins.
Unauthorized actions	Role- and ownership-based authorization on every API call.
Injection and script attacks	Input validation, parameterized queries, output encoding.
Fake or misleading listings	Required photos and fields; user reports; moderation queue.

Threat	Proposed mitigation
Personal data exposure	Data minimization; contact details only through in-app chat; HTTPS.
Unsafe meetups	Safe-meetup guidance; campus public locations suggested.
Payment fraud	Prefer in-person settlement for the MVP; escrow and a payment provider treated as future work.
Fake reviews	Reviews only from transaction participants; report option.

X. SUSTAINABILITY AND SOCIAL IMPACT

By giving reusable items a convenient route to the next student, Campus Connect is expected to encourage reuse and reduce unnecessary disposal, in line with the reuse emphasis of circular-economy definitions [3]. Students may save money and gain access to affordable resources, especially incoming students who need textbooks and basic furnishings, and renting reduces the need to purchase items needed only for a limited period. Motivations for participation in collaborative consumption include economic benefit and sustainability [2], but the strength of such effects in a campus setting must be measured, not assumed. No quantitative environmental impact is claimed; a sustainability impact tracker is a future extension.

XI. COMPARATIVE ANALYSIS

The project presentation compares Campus Connect with Facebook Marketplace, OLX/Craigslist, Mercari, and Depop. Feature and fee details of third-party platforms vary by region and change over time, and the presentation's platform-specific claims (for example, fee levels) were not independently verified for this draft; they are therefore omitted. Table VII compares design properties of a campus-restricted marketplace with those of a general-purpose marketplace, which is open to any registrant. Entries describe design intent and are not measured results.

TABLE VII
COMPARATIVE ANALYSIS (DESIGN-LEVEL)

Feature	Campus Connect	Generic marketplace	Advantage
University verification	University e-mail verification (described)	Open registration	Bounded, identifiable community
Student-specific categories	Described	Generic categories	Better fit to stud nt needs
Campus-local transactions	Meet on campus (described)	Any location	Shorter, safer meetups
Buy/sell/rent/ exchange	All four modes (described)	Varies by platform	Rental and swap support
Chat	In-app (described)	Varies by platform	Chat tied to listi ng
Ratings	Described	Varies by platform	Reputation withi n community
Student community	Campus members only	Mixed public	Shared context a d accountability
Shipping requirement	None for campus handover	Often shipping-based	Lower cost and delay

XII. METHODOLOGY

The methodology below is the **proposed** development process; the project material does not document the process actually followed. An iterative and incremental approach [17] is proposed because requirements are expected to be refined through feedback from prospective users (Fig. 6). **1) Requirement gathering:** interviews and surveys of students, faculty, and staff. **2) Requirement analysis:** classification, prioritization, and the SRS of Section V. **3) System design:** architecture of Section VI and interface prototypes. **4) Database design:** the logical model of Section VIII. **5) Frontend development:** React web interface and mobile screens. **6) Backend development:** API, verification, and messaging services. **7) Integration:** connection of clients, services, and database. **8) Testing:** the framework of Section XIII. **9) Deployment:** staged release to a pilot group within one campus. **10) Evaluation:** measurement against the expected outcomes of Section XIV, followed by the next iteration.

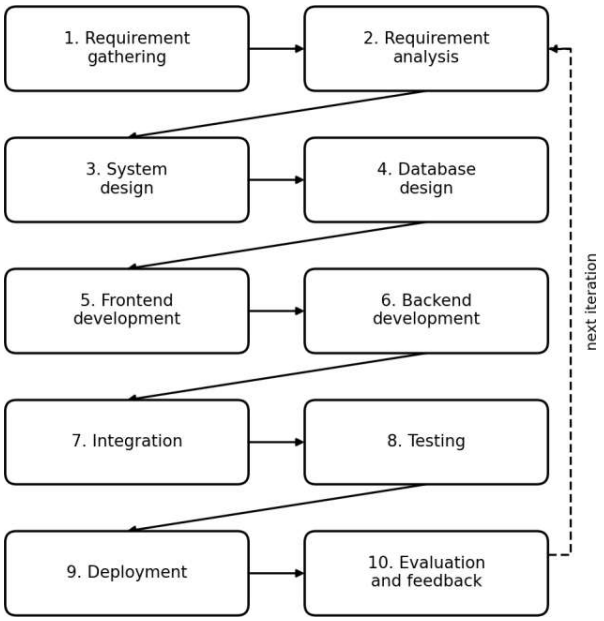


Fig. 6. Proposed iterative development methodology.

XIII. TESTING AND EVALUATION

A test framework is proposed in three groups. **Functional testing** covers registration, e-mail verification, login, listing creation, search, chat, wishlist, reviews, and notifications. **Security testing** covers authentication, authorization, input validation, session management, and data protection, guided by [11]. **Usability testing** covers navigation, the search experience, listing creation, and mobile responsiveness. Table VIII gives sample test cases. No test has been executed; the status column therefore reads "To be tested".

TABLE VIII
PROPOSED TEST CASES

Test ID	Test case	Input	Expected result	Status
TC-01	Registration with university e-mail	Valid name, university e-mail, password	Unverified account created; verification message sent	To be tested
TC-02	Registration with non-university e-mail	E-mail from another domain	Registration rejected with message	To be tested
TC-03	Verification with expired link	Expired code	Verification refused; new code can be requested	To be tested
TC-04	Login with valid credentials	Verified user's e-mail and password	Session started	To be tested
TC-05	Repeated failed logins	Several wrong passwords	Attempts throttled; no access granted	To be tested

Test ID	Test case	Input	Expected result	Status
TC-06	Create listing with photos	Required fields, three photos	Listing published and visible in category	To be tested
TC-07	Create listing with missing fields	Empty title	Validation error; no listing created	To be tested
TC-08	Search with keyword and category filter	“book”, category Books	Only matching available listings shown	To be tested
TC-09	Chat message delivery	Message from buyer to owner	Message stored and shown to recipient; unread indicator	To be tested
TC-10	Wishlist add and remove	Select listing twice	Listing added, then removed	To be tested
TC-11	Review by non-participant	Review request for others’ transaction	Request rejected (authorization)	To be tested
TC-12	Notification on new message	Message to offline user	Notification created for recipient	To be tested
TC-13	Injection input in search	Query containing query-operator characters	Input handled safely; no data disclosure	To be tested
TC-14	Session expiry and logout	Idle session; logout	Access to protected pages denied afterwards	To be tested
TC-15	Mobile layout navigation	Core tasks on a phone screen	Tasks completed without layout defects	To be tested

XIV. EXPECTED RESULTS AND DISCUSSION

No implementation or usage results are provided in the project material; the following are **expected outcomes**, to be tested as hypotheses in a pilot, and not measured results. **H1:** campus-local discovery becomes easier than on general platforms, which could be examined through time-to-find measures. **H2:** university verification improves perceived user trustworthiness, which could be examined through trust questionnaires. **H3:** peer-to-peer transactions become more convenient because listing, chat, and records are in one place. **H4:** more student resources are reused, which could be examined through counts of completed transactions. **H5:** ratings and reviews increase trust, subject to the limitations of reputation systems [4]. **H6:** campus-community interaction increases through events and errand requests. The pilot should define success thresholds before data collection.

XV. LIMITATIONS

Several limitations apply. Verification depends on university e-mail or identity systems and, as noted, does not prove real-world identity. The initial user base is limited to one campus, and a marketplace needs enough participants to be useful. Ratings cannot guarantee trust [4]. Payment security is unresolved in the described system and requires a provider or escrow design. Moderation effort is required to handle fake listings and disputes. Cooperation of the university may be needed for verification and event data. Scaling to multiple universities requires data separation and administration, and personal-data handling raises privacy obligations. Adoption by students is uncertain and has to be studied. Finally, the specification has not been validated by users, the literature review is not exhaustive, and the related student reports [9], [10] are not peer reviewed.

XVI. FUTURE WORK

Future work includes AI-based price recommendation, building on existing research on price prediction for second-hand items [12], [13], and a recommendation system for listings [14]; fraud and anomaly detection; scheduled pickup and delivery; secure escrow payments; a sustainability impact tracker; a loyalty and reward system; university ID integration; expansion to multiple campuses; a full mobile application; and advanced analytics. Empirical evaluation of the web MVP through the proposed test framework and a pilot study is the most immediate next step.

XVII. CONCLUSION

This paper addressed the lack of a trusted, campus-restricted marketplace for second-hand items. It proposed Campus Connect, a university-verified platform supporting buying, selling, renting, and exchanging, with chat, ratings, wishlists, notifications, campus events, and errand requests. The main contribution is a requirements-driven design: an SRS organized according to IEEE Std 830-1998, a layered architecture, use cases, a logical data model, a security analysis, a development methodology, and a test framework, with described, proposed, and future elements kept distinct. University verification is the central design element because it defines the community within which trust, convenience, and reuse can be pursued. The potential impact is expected, not demonstrated. Future research should evaluate the system through a pilot, strengthen verification and payment security, and examine multi-campus operation.

XVIII. COMPLIANCE WITH IEEE STD 830-1998

The requirements portion is mapped to the standard as follows: Introduction → Section V-A; Overall Description → Section V-B; Specific Requirements → Section V-C (Tables II and III). The qualities recommended by the standard are addressed as follows. *Correct*: each requirement is labelled by its source status (D, P, C). *Unambiguous*: each requirement has stated input, processing, and output, and open items are marked TBD. *Complete*: the described functions are covered, but TBD items (for example, performance thresholds and payment mechanism) remain. *Consistent*: terminology is defined in Section V-A. *Verifiable*: test cases are proposed in Table VIII. *Prioritized*: High, Medium, and Low priorities are assigned. *Traceable*: identifiers link use cases to requirements (Section VII). *Modifiable*: requirements are uniquely numbered and kept separate from design. This is a draft specification and no formal certification or complete compliance with the standard is claimed.

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